Ain Shams University
Faculty of Engineering
Electronics and Communications Department
Senior-1
Spring 2026

ECE353s – Wireless Communication Networks

Project

(A) It is required to design using MATLAB/Python, a simple planning tool for a service provider that owns 340 channels in the 900 MHz band. Your code should ask for the GOS, city area, user density, SIR_{min}, and sectorization method. Assume blocked calls are cleared in this system. Then, it should produce the following design parameters:

1) Cluster Size.
2) Total number of cells in city.
3) Cell radius.
4) Traffic intensity per cell and traffic intensity per sector.
5) Base station transmitted power.
6) A plot for the MS received power in dBm versus its distance from the BS.

In your design, Use Hata model (as outdoor propagation channel model) while assuming urban-medium city. Let, the traffic per user equals 0.025 Erlang, the effective heights of BS and MS equal 20 and 1.5 meters respectively, and MS sensitivity equals $-95\ dB$, and the path loss exponent equals 4.

Note: You can find Erlang B table in the second attachment.

(B) To validate your planning tool and understand the trade-offs between different design parameters it is required to deliver for a city of area equals 100 Km^2 the following figures with reasonable comments. Each figure should contain three curves for omni-directional, 120° sectorization and 60° sectorization designs.

1) A plot for the cluster size versus $SIR_{min}(1dB\ to\ 30\ dB)$.

2) At SIR_{min}=$19dB$ and user density = 1400 users/km^2.
   (i) A plot for the number of cells versus GOS (1% to 30%).
   (ii) A plot for the traffic intensity per cell versus GOS (1% to 30%).

3) At SIR_{min}=$14dB$ & user density = 1400 users/km^2.
   (i) A plot for the number of cells versus GOS (1% to 30%).
   (ii) A plot for the traffic intensity per cell versus GOS (1% to 30%).

4) At SIR_{min} = 14dB & GOS = 2%
   (i) Plot the number of cells versus user density (100 to 2000 users/km^2).
   (ii) Plot the cell radius versus user density (100 to 2000 users/km^2).

5) At SIR_{min} = 19dB & GOS = 2%
   (i) Plot the number of cells versus user density (100 to 2000 users/km^2).
   (ii) Plot the cell radius versus user density (100 to 2000 users/km^2).

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Note:
- You need to illustrate the design criteria you selected to use and the parameter/parameters you selected to optimize in your design. You can assume any missing parameters (**if only needed**).
- Your code should be generic to be used for different given input values as described in part (**A**).

Deliverables:

Please submit a soft-copy of the following items:
- All MATLAB/Python codes that produce (**A**) and (**B**).
- A short report explaining the procedure you used along with the mathematical equations. The report should also include all the figures generated by your code and your comments on each figure.
- The report should include explanation for Hata model (Equation, pros & cons compared to studied models, the effects it considers, and the effects it neglects)

Project rules:

- Group project (4-5 students/group).
- No late reports are accepted.
- If 2 reports or codes are similar, both will have zero marks.
- If a project is copied from Internet. It will have zero mark.
- Project discussion method "To be determined later"
- Delivery method "On LMS" as separate files not compressed files.
- Project report must contain the names and IDs of all group members
- Only one member in the group must submit his/her project on LMS.